

Question Paper

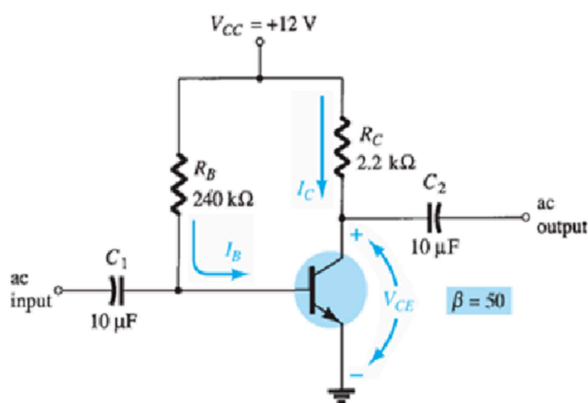
1470 - ISA 2 - DEC 2025 - UE25EC141A - (GSet-1)

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1.a : Marks (4.0)

Determine the following for the fixed-bias configuration of the given circuit.

1. I_B
2. I_C
3. V_{CE}
4. V_{BC}

**Expected Answer**

$$\text{a) } I_{BQ} = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 \text{ V} - 0.7 \text{ V}}{240 \text{ k}\Omega} = 47.08 \mu\text{A}$$

$$\text{b) } I_{CQ} = \beta I_{BQ} = (50)(47.08 \mu\text{A}) = 2.35 \text{ mA}$$

$$\begin{aligned} \text{c) } V_{CEQ} &= V_{CC} - I_C R_C \\ &= 12 \text{ V} - (2.35 \text{ mA})(2.2 \text{ k}\Omega) \\ &= 6.83 \text{ V} \end{aligned}$$

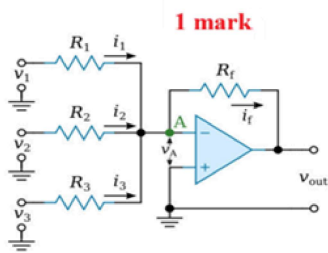
$$\begin{aligned} \text{d) } V_{BC} &= V_B - V_C = 0.7 \text{ V} - 6.83 \text{ V} \\ &= -6.13 \text{ V} \end{aligned}$$

1 marks each

1.b : Marks (4.0)

With neat circuit diagram, and necessary equations, explain the operational amplifier as adder.

Expected Answer



Applying KCL at node A,

$$i_1 + i_2 + i_3 = i_f$$

2 marks

Where,

$$i_1 = \frac{v_1 - v_A}{R_1}$$

$$i_3 = \frac{v_3 - v_A}{R_3}$$

$$i_2 = \frac{v_2 - v_A}{R_2}$$

$$i_f = \frac{v_A - v_{out}}{R_f}$$

Substituting eq (2) in (1) and considering $v_A = 0$ (Virtual ground),

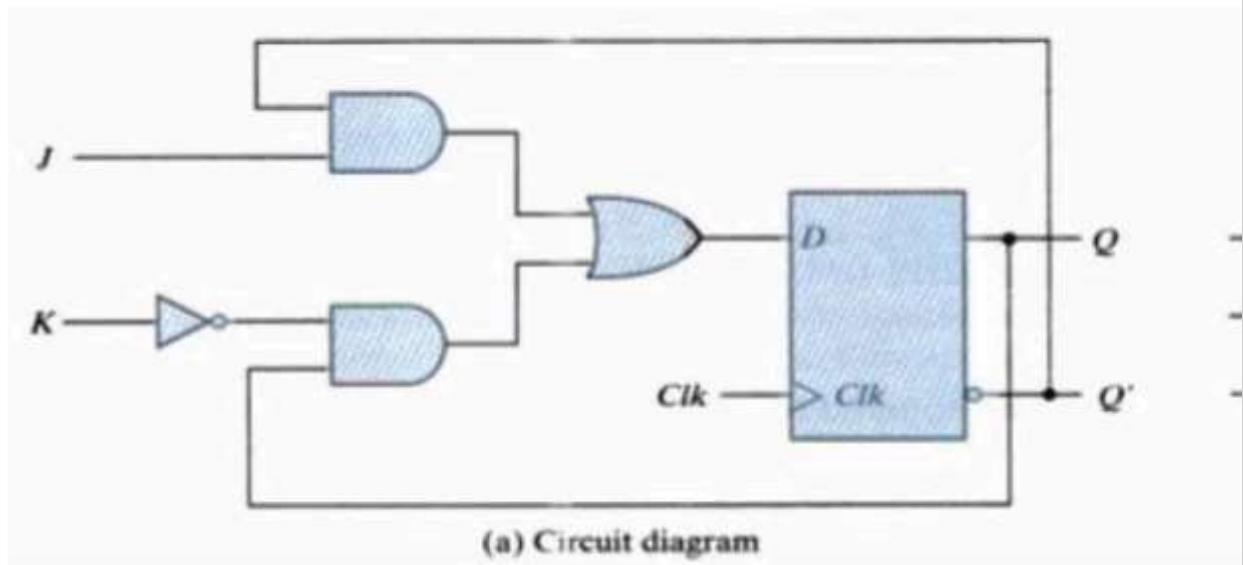
$$\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} = \frac{-v_{out}}{R_f} \quad \text{or} \quad v_{out} = -R_f \left(\frac{v_1}{R_1} + \frac{v_2}{R_2} + \frac{v_3}{R_3} \right) \quad \text{--- 1 mark}$$

1.c : Marks (4.0)

Explain J-K Flip flop using D flip flop with Circuit diagram and Function Table (1M + 1M+ 2M)

Expected Answer

Answer: Circuit diagram: 1 mark, Function Table 1 Mark and Explanati



Function Table

Flip-Flop Characteristic Tables

JK Flip-Flop

J	K	$Q(t + 1)$	
0	0	$Q(t)$	No change
0	1	0	Reset
1	0	1	Set
1	1	$Q'(t)$	Complement

1.d : Marks (4.0)

Describe Half Adder with Truth Table, Boolean Expression and Logic diagram using Basic Gates and NAND Gates (0.5M+0.5M+1M+2M)

Expected Answer



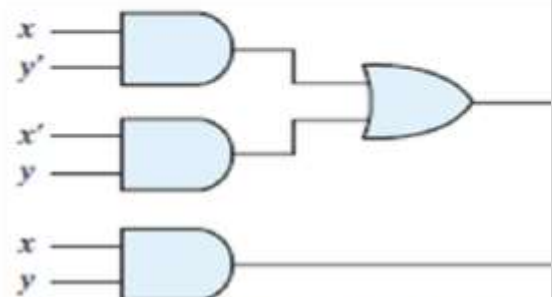
**Answer: (a) Truth Table - 0.5 marks (b) Boolean Expression: 0.5Marks
(c) Logic diagram using Basic gates: 1Mark and NAND Gates: 2 marks**

❖ Half Adder

- x and y are the two binary inputs
- Sum (s) and Carry (c) are the two binary outputs

Truth Table

x	y	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

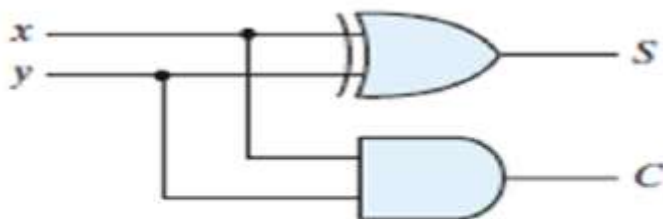


Logic Diagram

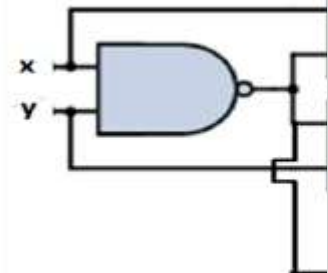
❖ **Half Adder:** $s = x \oplus y$
 $c = x.y$

Sum expression: s

❖ Half Adder using Logic Gates



❖ Half Adder u



5 : Marks (1.0)

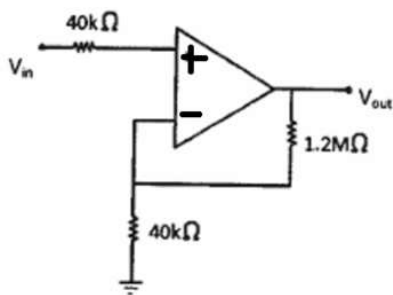
Choose the correct statement.

Expected Answer

An Operational Amplifier is a high gain differential amplifier with infinity input impedance and low output impedance.

6 : Marks (1.0)

In the Op-Amp circuit shown below, what is the input voltage if output voltage is -1V.



Expected Answer

$v_{in} = V_o / \text{gain}$, gain of the circuit is 31
 $v_{in} = -1/31 = -0.0322 = -32.2\text{mV}$

7 : Marks (1.0)

For the transistor in CB - input characteristics, in active region if the value of V_{CB} increases then

Expected Answer

The Emitter current increases

8 : Marks (1.0)

In a common base connection, the emitter current is 1mA. If the emitter circuit is open, the collector current is 50 μ A. Given that $\alpha = 0.92$, the total collector current is

Expected Answer

0.97mA

$$I_c = \alpha \times I_E + I_{CBO}$$

9 : Marks (1.0)

The _____ part of the transistor is heavily doped

Expected Answer

Emitter

10 : Marks (1.0)

In operational amplifier, the ratio of differential gain to the common mode gain is known as

Expected Answer

CMRR -Common mode rejection ratio

11 : Marks (2.0)

A transistor has $I_B = 105\mu\text{A}$ and $I_C = 2.05\text{mA}$ find

- The current amplification factor
- If the I_B changes by $+50\mu\text{A}$ and I_C changes by $+1.5\text{mA}$, then the new current amplification factor is

Expected Answer

a = 19.5 and b = 22.9

- The current amplification factor is beta, $I_C / I_B = 19.5$
- Beta = $3.55\text{mA} / 155\mu\text{A} = 22.9$

12 : Marks (1.0)

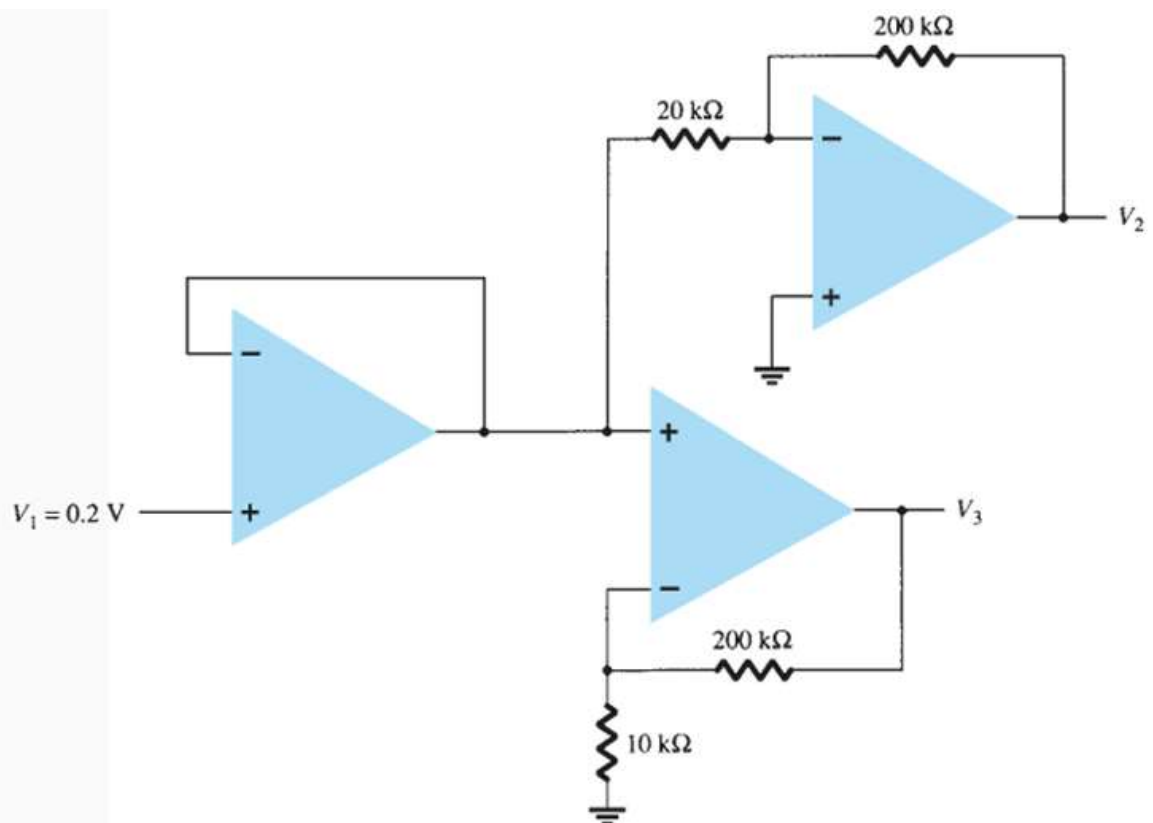
The region of operation of the transistor, when the ***emitter base junction (EB Jn)*** in forward bias and ***collector base junction (CB Jn)*** in reverse bias condition is

Expected Answer

Active / linear region

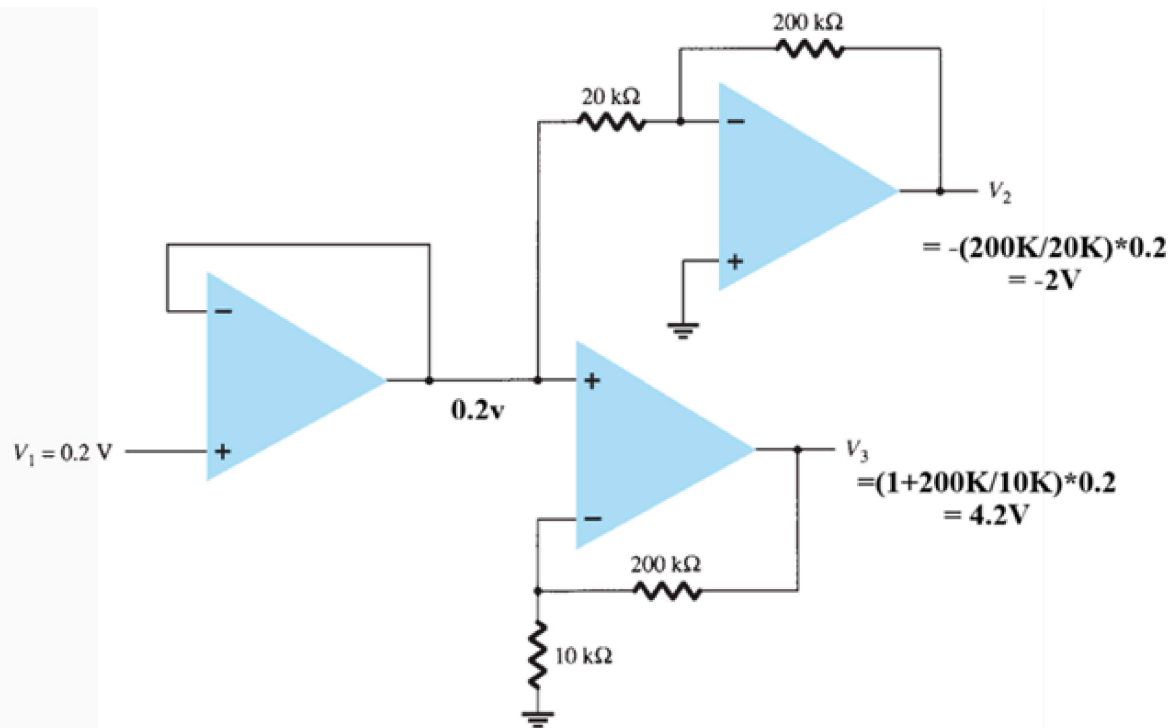
13 : Marks (2.0)

The output voltages V_2 and V_3 in the circuit of Figure below is



Expected Answer

$$V_2 = -2V, V_3 = 4.2V$$



14 : Marks (1.0)

NPN transistor are preferred over PNP transistors because

Expected Answer

NPN have higher mobility of electrons than the mobility of holes

15 : Marks (2.0)

The simplification of Boolean expression $F = A.B.C + A' + A.B'.C$ will result in

Expected Answer

Solution: $F = A.C (B + B') + A'$ where $B + B' = 1$

$$F = A.C + A'$$

$F = (A' + A). (A' + C)$ Distributive Law: $A + B.C = (A+B).C$

$$F = A' + C$$

16 : Marks (1.0)

The Function $F(A,B,C) = m_3 + m_4$ represents the following SOP Logical expression

Expected Answer

$F = A\bar{B}C + AB\bar{C}$
($m_3 - 011$ and $m_4 - 100$)

17 : Marks (1.0)

When logic signal A and its exact inverse $\bar{A}$ is connected to the inputs of EXOR Gate, then the output of the EX-OR gate if $A=1$

Expected Answer

Answer : If $A = 1$ and $A' = 0$ then output of EXOR gate = $A \text{ XOR } A' = 1$

18 : Marks (1.0)

The T Flip flop is designed by _____

Expected Answer

Answer : Shorting J and K

19 : Marks (1.0)

Which of the following is the correct Boolean expression?

Expected Answer

$A + A' = 1$

20 : Marks (1.0)

The standard canonical sum of the product form of the function $F(A,B) = A+B$

Expected Answer

Solution: $F(A,B) = A+B$

$$= A(B+B?) + B(A+A?)$$

$$= AB + AB? + AB + A?B$$

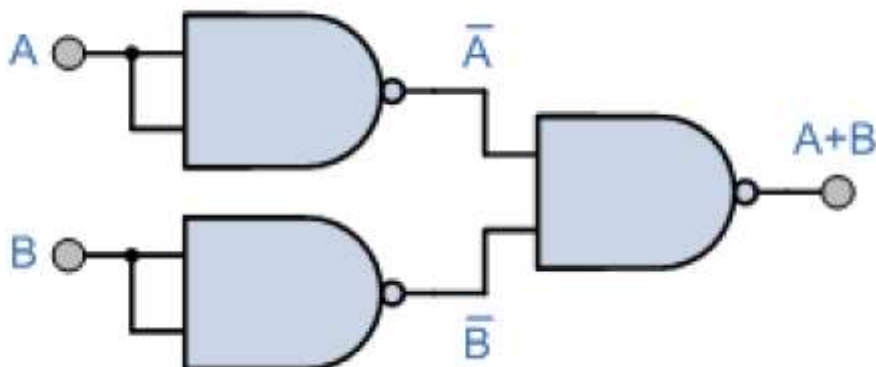
$$= AB + AB? + A?B$$

21 : Marks (1.0)

How many minimum NAND gates are required to realize OR Gate?

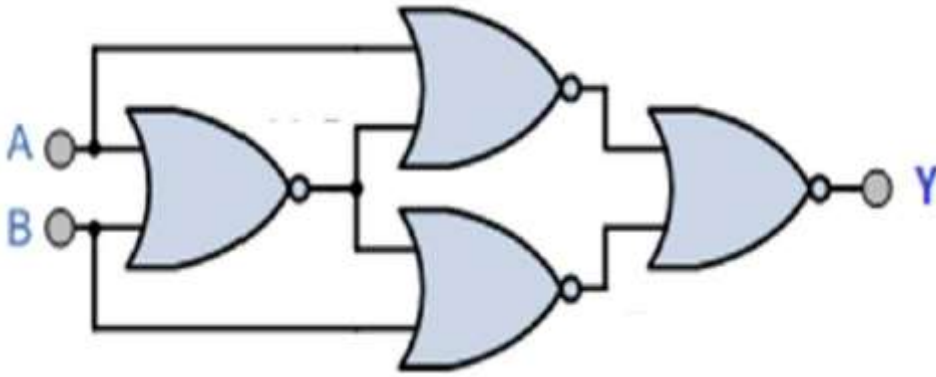
Expected Answer

Answer : 3



22 : Marks (1.0)

The output of the following circuit is _____

**Expected Answer**

$Y = A \oplus B$, XNOR Gate

23 : Marks (1.0)

Subtract $(1010)_2$ from $(1111)_2$ using 2's complement method.

Expected Answer

Answer : 0101

$$\begin{array}{r} 1111 \\ + 0110 \\ \hline 10101 \end{array}$$

Omit this carry (1)

↓

0101 *Answer*

24 : Marks (2.0)

The Canonical POS Form for f_1 and f_2

x	y	z	Function f_1	Function f_2
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Expected Answer

Solution:

❖ Canonical POS Form for

$$\begin{aligned}
 f_1 &= (x + y + z)(x + y' + z)(x + y' + z')(x' + y' + z) \\
 &= M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6
 \end{aligned}$$

❖ Canonical POS Form for f_2

$$\begin{aligned}
 f_2 &= (x + y + z)(x + y + z')(x + y' + z)(x' + y' + z) \\
 &= M_0 \cdot M_1 \cdot M_2 \cdot M_4
 \end{aligned}$$